

Clémence SEBE

Third-year doctoral student at LISN,
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UNIVERSITY DEGREES

Master's Degree in Computational Biology: Analysis, Modeling, and Engineering of Biological and Medical Information 2021 - 2023
SaclAI-School Excellence Scholarship

With Honours

UNIVERSITÉ PARIS-SACLAY

Expertise in bioinformatics through RNA-seq, sequence and protein structure analysis, statistical and image analysis, web application development, and the design of reproducible workflows using Nextflow

Double Bachelor's Degree in Computer Science and Mathematics 2018 - 2021

With Honours

UNIVERSITÉ PARIS-SACLAY

RESEARCH EXPERIENCE

PhD Student 2023 -

LISN, UNIVERSITÉ PARIS-SACLAY

Thesis title: Information Extraction from Scientific Literature: A Case Study on Bioinformatics Workflows

Supervised by Sarah Cohen-Boulakia, Olivier Ferret, and Aurélie Névéol

- Designed and coordinated annotation campaigns to create annotated corpora for named entity recognition in bioinformatics workflow publications and executable Nextflow process code [1,4]. Developed a gold-standard dataset for entity resolution.
- Trained encoder- and decoder-based (LLM) models for named entity recognition. Extended model vocabularies and fine-tuned language models to improve domain-specific information extraction [1,4].
- Analyzed bioinformatics tool knowledge bases to assess their coverage, structure, and interoperability.
- *Teaching Assistant (64h/year)* : Advanced Functional Programming, Introduction to Data Science.

Doctoral Stay 2025

HUMBOLDT-UNIVERSITÄT, BERLIN

Three-month internship within the Knowledge Management in Bioinformatics team

Supervised by Ulf Leser

- Working on entity resolution using both supervised and unsupervised approaches, including word embedding similarity and knowledge-base bridging techniques [1].

Research Intern 2022-2023

LISN, UNIVERSITÉ PARIS-SACLAY (3 and 5 months)

Extracting Information on Scientific Workflows from Literature and Concordance between the description of bioinformatics workflows in scientific publications and their implementation in the associated code

- Designed and coordinated an annotation campaign to create a richly annotated corpus of scientific publications on bioinformatics workflows, covering 16 entity types [4,8].
- Conducted initial experiments on methods for entity resolution across two heterogeneous data sources.

Study and Research Project

2021-2022

LISN, UNIVERSITÉ PARIS-SACLAY (9 months, 2 days per week)

Analysis of Processes in Nextflow Workflows and

Modeling and Implementation of a Database Compiling All Information on Workflows

- Extracted structural representation of Nextflow workflow processes.
- Designed and implemented a *PostgreSQL* database integrating both high-level meta-data and process-level code information from multiple Nextflow workflows

CO-SUPERVISION OF INTERNS

- 2026 Sukhjot Kaur (M1 - Université Paris-Saclay) : **Development of a Web Interface for the CoPaLink Tool**, with Sarah Cohen-Boulakia, Olivier Ferret, Aurélie Névéol.
- 2025 Anjali Badlani (M2 - University of Potsdam) : **NLP-based Workflow Analysis in Snake-make Repositories**, with Sarah Cohen-Boulakia.

PUBLICATIONS

International Journals

- [1] **Sebe Clémence**, Olivier Ferret, Aurélie Névéol, Mahdi Esmailoghli, Ulf Leser, and Sarah Cohen-Boulakia. Supporting Workflow Reproducibility by Linking Bioinformatics Tools across Papers and Executable Code, mars 2026. arXiv:2603.08195 [cs], under review for *Bioinformatics*
- [2] Marine Djaffardjy, George Marchment, **Sebe Clémence**, Raphaël Blanchet, Khalid Belhajjame, Alban Gaignard, Frédéric Lemoine, and Sarah Cohen-Boulakia. Developing and reusing bioinformatics data analysis pipelines using scientific workflow systems. *Computational and Structural Biotechnology Journal*, 2023

Book Chapter

- [3] Sarah Cohen-Boulakia, Frédéric Lemoine, George Marchment, Marine Djaffardjy, Alban Gaignard, **Sebe Clémence**, and Khalid Belhajjame. Reuse and reproduce bioinformatic pipelines using scientific workflow systems. In *Workflow systems for large-scale scientific data analysis*. Berlin Universities Publishing, 2026

International Peer-Reviewed Conferences

- [4] **Sebe Clémence**, Sarah Cohen-Boulakia, Olivier Ferret, and Aurélie Névéol. Extracting Information in a Low-Resource Setting: Case Study on Bioinformatics Workflows. In *Advances in Intelligent Data Analysis (IDA) XXIII*, 2025

Poster

- [5] George Marchment, Marie Schmit, **Sebe Clémence**, Frédéric Lemoine, Hervé Ménager, and Sarah Cohen-Boulakia. Representing bioinformatics Nextflow workflows in RO-Crate : challenges and opportunities. SWAT4HCLS, 2024

- [6] **Sebe Clémence**, Frédéric Lemoine, Alban Gaignard, Olivier Ferret, Sarah Cohen-Boulakia, and Aurélie Névél. Towards improving workflows reproducibility : Extracting information on workflows from text and code repositories. ISMB/ECCB, 2023

National Peer-Reviewed Conferences

- [7] Marie Schmit, Melvin Selim Atay, Khalid Belhajjame, Ulysse Le Clanche, Emmanuel Coquery, Olivier Dameron, Fabien Duchateau, Alban Gaignard, Mouna El Garb, Jaffar Gura, Nicolas Lumineau, George Marchment, Camille Maumet, **Sebe Clémence**, Frédéric Lemoine, Sarah Cohen-Boulakia, and Hervé Ménager. ShareFAIR-KG, a centralised knowledge base of scientific workflows. In *JOBIM*, 2026
- [8] **Sebe Clémence**, Sarah Cohen-Boulakia, Olivier Ferret, and Aurélie Névél. Extraction d’entités nommées décrivant des chaînes de traitement bioinformatiques dans des articles scientifiques en anglais. In *TALN*, 2024
- [9] **Sebe Clémence**, Aurélie Névél, Sarah Cohen-Boulakia, and Alban Gaignard. Extraction d’informations sur les workflows scientifiques à partir de la littérature. In *Extraction et Gestion des Connaissances, RNTI-E-39*, 2023

DEVELOPED TOOLS AND RESOURCES

CoPaLink

CoPaLink is a tool designed to automatically extract the names of bioinformatics tools mentioned in a scientific article describing a Nextflow workflow, as well as in the executable source code of the workflow. A linking step between the bioinformatics tool names from both sources is then performed.

GitLab repository: <https://gitlab.liris.cnrs.fr/sharefair/copalink>

COMPUTER SKILLS

Programming Languages Mastered	: PYTHON, SQL, BASH, L ^A T _E X.
Other Known Languages	: C, JAVA, OCAML.
Databases	: PostgreSQL.
Other Tools	: R, Github.

OTHER RESEARCH-RELATED ACTIVITIES

PhD Student Representative at LISN	2024-2025
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Participation in the laboratory council, organization of events for PhD students

(Invited) Oral Presentation at the Scientific Day of the Bioinformatics Axis of Biogenouest	2024
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Title: Extraction d’informations sur les workflows scientifiques à partir de la littérature anglaise: premiers résultats, May 24, Rennes

Oral Presentation during <i>ShareFAIR</i> Project Days	2023 - 2026
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- *Extracting and Linking Workflow Information between Bioinformatics Literature and Source Code*, December 10-11, 2025, Institut Pasteur
- *BioToFlow corpus*, November 28-29, 2024, Institut Pasteur
- *Extraction de noms d’outils bioinformatiques dans la littérature scientifique*, November 20-21, 2023, Université Paris-Saclay

Oral Presentation during Team Seminars at LISN 2023-2026

- *Bioinformatics Team*, January 12, 2023, and February 5, 2026
- *Reading Group of the Language Sciences and Technologies Department*, September 27, 2024

Oral Presentation at the Junior Multidisciplinary Congress of Université Paris-Saclay 2022-2023

- *Extraction d'Informations sur les Workflows Scientifiques à partir d'Articles en Anglais*, June 8, 2023, ENS Paris-Saclay
- *Réutilisation de Workflows d'Analyse Bioinformatique*, June 2, 2022, ENS Paris-Saclay

Oral Presentation at R2P2 (*Analyse de données multi-modales pour les Pathologies complexes par la conception et l'implémentation de Protocoles Reproductibles et Réutilisables*) Project Days 2021

OUTREACH ACTIVITIES

Film-Debate 2026

Participation in a Film-Debate with high school students from Lycée Blaise Pascal, Orsay on April 2. Film: "Le Théorème de Marguerite"

Speed-Meeting 2026

- *Ambition Sciences pour elles*, mentoring program with high school girls from Lycée Marceau in Chartres, March 12
- *Career guidance for gender equality*, with 9th-grade students from Collège Charles Péguy, Palaiseau, December 15